

G. W. Leibniz DE ANALYSI SITUS

Translated by David Jekel and Matthew McMillan

Last revision: 28 Aug 2026

Manuscript: https://digitale-sammlungen.gwlb.de/index.php?id=6&tx_dlf%5Bid%5D=1642&tx_dlf%5Bpage%5D=1

Typescript: <https://www.gwlb.de/fileadmin/Leibniz/repositorium-des-leibniz-archivs/LAA-Philiumm-v3.pdf#page=122>

Leibniz Katalog Nr. 40852: <https://leibniz-katalog.bbaw.de/en/extended-search?katnr=40852>

PHILIUMM. Transkriptionen und Vorauseditionen mathematischer Schriften für die Leibniz-Akademie-Ausgabe. Version 3, N. 25, S. 112–116. 25. November 2025.

De Analysis Situs

[1] Quae vulgo celebratur *Analysis mathematica*, est *magnitudinis*, non *situs*; atque adeo directe quidem et immediate ad Arithmeticam pertinet, ad Geometriam autem per circuitum
5 quendam applicatur. Unde fit ut multa ex consideratione situs facile pateant, quae calculus Algebraicus aegrius ostendit. Problemata Geometrica ad Algebra, id est quae figuris determinantur ad aequationes revocare res non raro
10 satis prolixa est, et rursus alia prolixitate difficultateque opus est, ut ab aequatione ad constructionem, ab Algebra ad Geometriam redeatur, saepeque hac via non admodum aptae prodeunt constructiones nisi feliciter in quasdam non
praevisas suppositiones assumptionesve incidamus. Hoc ipse
15 Cartesius tacite fassus est, cum lib. 3. Geometriae suae problema quoddam Pappi resolvit. Et sane Algebra sive numerica sive speciosa, addit, subtrahit, multiplicat, dividit, radices extrahit, quod utique arithmeticum est. Nam ipsa Logistica, seu scientia magnitudinis proportionisve in universum, nihil aliud tractat quam numerum generalem seu indeterminatum,
20 et has in eo species operandi quoniam *Magnitudo* revera determinatarum partium multitudine aestimatur, quae tamen manente re variat, prout alia aut alia Mensura vel Unitas assumitur. Unde mirum non est, Scientiam Magnitudinis in universum esse Arithmeticae genus, cum agat de numero

De Analysis Situs

[1] What is commonly lauded as *mathematical analysis* is analysis of *magnitude*, not of *situation*, and thus it is directly and immediately apposite to arithmetic, but to geometry it applies in a certain round-
5 about fashion. And so it happens that by considering situation, many things easily become clear which the algebraic calculus shows with greater difficulty. To reduce geometrical problems to algebra, that is, to reduce what is determined by figures to equations, is often a rather lengthy process; and yet another protracted and arduous effort
10 is needed to return from equation to construction, from algebra to geometry; and often by this route we get constructions that are not quite suitable, unless we are fortunate enough to strike upon certain unforeseen suppositions or assumptions. Descartes tacitly admits this very thing when he solves a certain problem of Pappus in the
15 third book of his Geometry. And indeed algebra, whether numeric or symbolic [*speciosa*], adds, subtracts, multiplies, divides, and extracts roots, which is, of course, arithmetic. For logistic itself, i.e. the science of magnitude and proportion in general, handles nothing but general or indeterminate number, and the symbols of operations on it, since
20 *magnitude* is actually assessed by the multiplicity of determinate parts, which nevertheless varies according as one or another measure or unit is assumed, while the thing remains the same. Hence it is not surprising that the science of magnitudes in general is a kind of arithmetic, since it deals with uncertain numbers.

incerto.

[2] Habebant Veteres aliud Analyseos genus, ab Algebra diver-
sum, quod magis ad situs considerationem accedit, tractans
de *Datis* et de *Sedibus* quaesitorum seu *Locis*. Et huc tendit
5 Euclidis libellus de *Datis*, in quem Marini Commentarius
extat. De *Locis* vero planis, solidis, Linearibus actum est cum
ab aliis, tum ab Apollonio, cujus propositiones Pappus con-
servavit, unde recentiores Loca plana solidaque restituerunt.
10 Sed ita, ut veritatem magis quam fontem doctrinae veteris
ostendisse videantur. Hoc tamen Analyseos genus neque ad
calculus rem revocat, neque etiam producit usque ad prima
principia atque Elementa situs, quod ad perfectam Analysin
necesse est.

[3] Vera igitur Situs Analysis adhuc supplenda est, idque
15 vel ex eo constat, quod omnes Analytici, sive Algebram ex-
erceant novo more, sive data et quaesita ad veterem formam
tractent, multa ex Geometria elementari assumere debent,
quae non ex magnitudinis sed figurae consideratione dedu-
cuntur neque determinata quadam via hactenus patent. Eu-
20 clides ipse quaedam axiomata satis obscura sine probatione
assumere coactus est, ut caetera procederent. Et Theore-
matum Demonstratio solutioque problematum in Elementis
magis aliquando apparet, laboris opus quam methodi et ar-
tis, quanquam et interdum artificium processus suppressum
25 videatur.

[4] Figura in universum praeter quantitatem continet quali-
tatem seu formam; et quemadmodum *aequalia* sunt quorum
eadem est magnitudo, ita *similia* sunt quorum eadem est
forma. Et similitudinum quidem seu formarum consideratio
30 latius patet quam mathesis, et ex Metaphysica reperitur, sed
tamen in Mathesi quoque multiplicem usum habet, inque

[2] The ancients had another kind of analysis, distinct from algebra,
which comes nearer to the consideration of situation, treating of *givens*
and *sites*¹ or *loci* of things sought. And Euclid's little book about givens,
5 on which we have the commentary of Marinus, aims at this. Planar,
solid, and linear loci were of course dealt with by others, and also
by Apollonius, whose propositions were preserved by Pappus, from
which more recently some have recovered the planar and solid loci,
but so that they seem to have shown the truth more than the source of
10 the ancient doctrine. This kind of analysis, however, neither reduces
the matter to calculus, nor indeed is it carried all the way to the first
principles and Elements of situation, which is necessary for a perfect
analysis.

[3] Therefore, the true analysis of situation is yet to be supplied, and
this is evident from the fact that all analysts either practise algebra in
15 the new fashion or treat the givens and things sought according to the
ancient form, and they must assume many things from elementary
geometry which are not deduced from consideration of magnitude
but of figure, and neither are they clear by any determinate path,
even to this day. Euclid himself was compelled to assume certain
20 quite obscure axioms without proof, that the rest might proceed. And
the demonstration of theorems and the solution of problems in the
Elements sometimes appears to be a work more of toil than of method
and art, although now and then an art to the process seems to be
concealed.

[4] Figure in general includes, besides quantity, also quality or form;
25 and just as *equals* are things whose magnitude is the same, so *similars*
are things whose form is the same. And indeed, the consideration of
similarities or forms is clearly broader than mathesis and is learned
from metaphysics, though it does also have many and varied uses in
30 mathesis and is beneficial for the algebraic calculus itself; but similar-

¹Latin *sedibus*, literally 'seats', is not common in mathematical writing. It means the place occupied or inhabited by a thing.

ipso Calculo Algebraico prodest, sed omnium maxime similitudo spectatur in sitibus seu figuris Geometriae. Itaque Analysis vere Geometrica non tantum aequalitates spectat et proportionalitates, quae revera ad aequalitates reducuntur; sed similitudines etiam, et ex aequalitate ac similitudine conjunctis natas congruentias adhibere debet.

[5] Causam vero cur similitudinis consideratione non satis usi sunt Geometrae, hanc esse arbitror, quod nullam ejus notionem generalem haberent satis distinctam aut ad Mathematicas disquisitiones accommodatam, vitio philosophorum, qui definitionibus vagis et definito obscuritate paribus, in prima praesertim philosophia contenti esse solent, unde mirum non est sterilem esse solere doctrinam illam et verbosam. Itaque non sufficit similia dicere, quorum eadem forma est, nisi *formae* rursus generalis notio habeatur. Comperi autem, instituta qualitatis vel formae explicatione rem tandem eo devenire, ut *similia* sint, quae singulatim observata discerni non possunt. Quantitas enim sola rerum comparientia seu applicatione actuali interveniente deprehendi potest, qualitas aliquid menti objicit, quod in re separatim agnoscas et ad comparisonem duarum rerum adhibere possis, actuali licet applicatione non interveniente, qua res rei vel immediate vel mediante tertio tanquam mensura confertur. Fingamus duo templa vel aedificia exstructa haberi, ea lege, ut nihil in uno deprehendi queat, quod non et in alio observetur: nempe materiam ubique eandem esse, marmor Parium candidum, si placet; parietum, columnarum, caeterorumque omnium easdem utrobique esse proportionem; angulos utrobique eosdem, seu ejusdem rationis ad rectum; itaque qui in haec bina templa ducetur clausis oculis, sed post ingressum apertis, et nunc in uno, nunc in altero versabitur, nullum indicium

ity is seen most of all in situations, i.e. figures of geometry. And so a truly geometric analysis regards not only equalities and proportions, which actually reduce to equalities, but indeed similarities, and it should employ the congruences arising from equality and similarity combined.

[5] Now the reason why geometers have not made satisfactory use of the consideration of similarity I judge to be this: that they had no general notion of it sufficiently distinct or adapted to mathematical inquiries, a fault of philosophers who habitually content themselves with vague definitions equal in obscurity to what is defined, especially in first philosophy; whence it is no wonder that the teaching tends to be sterile and verbose. And so it is not enough to say that similars are things whose form is the same unless one has in turn a general notion of *form*. Now I learned, while working out an explanation of quality or form, that in the end the matter comes down to this, that similars are things which cannot be distinguished when observed individually. For quantity can only be apprehended by the co-presence of things, i.e. with an actual application² intervening. Quality presents something to the mind which you recognize in a thing separately and can use for comparison of two things, even with no actual application intervening by which one thing is compared³ with another either immediately or by mediation of a third thing as a measure. Let us imagine that two temples or buildings have been constructed by the rule that nothing can be apprehended in the one that you would not also observe in the other: the material is of course everywhere the same, white marble of Paros if you like; the proportions of all the walls, the columns, and the rest are the same in both; the angles in both are the same, i.e. of the same ratio to a right angle; and so whoever is led into these two temples with eyes closed, and after entering opens them, and goes about now in the one and now in the other,

²Latin *applicatione actuali interveniente*; here *applicatio* is used as in the tradition of English geometry texts to render a term of Euclid that indicates a bringing of one thing to another by rigid motion; often the two are assumed to be congruent and *applicatio* results in superposition, though that is not the case here.

³Latin *confertur*. Although typically glossed as ‘compared’, it is worth pointing out that the Latin sense more evidently includes an element of ‘bringing together’ that is pertinent here.

ex ipsis inveniet, unde alterum ab altero discernat. Et tamen magnitudine differre possunt, atque adeo discerni poterunt si simul spectentur, ex loco eodem; vel etiam (licet remota sint invicem) si tertium aliquod translatus nunc cum uno, 5 nunc cum altero comparetur, veluti si mensura aliqua, qualis ulna aut pes aut aliud quiddam ad metiendum aptum, nunc uni nunc alteri accommodetur, nam tum demum discernendi ratio dabitur inaequalitate deprehensa. Idem est, si ipsum spectatoris corpus, aut membrum, quod utique cum ipso de 10 loco in locum transit mensuraeque officium praestat, his templis applicetur; tunc enim magnitudo diversa, et per hanc discernendi modus apparebit. Sed si spectatorem non nisi ut mentem oculatam consideres, tanquam in puncto constitutam, nec ullas secum magnitudines aut re aut imaginatione 15 afferentem, eaque sola in rebus considerantem, quae intellectu consequi licet, velut numeros, proportiones, angulos, discrimen nullum occurret. Similia igitur dicentur haec templa, quia non nisi hac coobservatione vel inter se, vel cum tertio, minime autem sigillatim et per se spectata discerni 20 potuere.

[6] Haec evidens et practica, et generalis similitudinis descriptio nobis ad demonstrationes Geometricas proderit, ut mox patebit. Nam duas figuras oblatas, similes dicemus, si aliquid in una singulatim spectata notari nequeat; quod in 25 altera non aequè deprehendatur. Itaque eandem utrobique ingredientium rationem sive proportionem esse debere consequitur, alioqui per se sigillatim, seu nulla licet amborum coobservatione instituta, discrimen apparebit. At Geometrae cum generali similitudinis notione carerent, figuras similes ex 30 aequalibus respondentibus angulis definierunt, quod speciale est, non ipsam naturam similitudinis in universum aperit. Itaque circuitu opus fuit, ut demonstrarentur, quae ex nostra notione primo intuitu patent. Sed ad exempla veniamus.

will not find any token in them by which to distinguish the one from the other. And yet they can differ in magnitude, and thus can be distinguished if they are viewed simultaneously, from the same place, or even (letting them be far from each other) if some third thing is transported and compared now with the one and now with the other, 5R just as some measure such as a cubit or foot or something else suitable for measuring is fitted now to the one and now to the other, for then at last a basis for distinguishing will be given by the observed inequality. It is the same if the viewer's body itself, or a limb, which certainly 10R goes with oneself from place to place and serves the role of a measure, is applied to these temples; for then the different magnitude, and through it a method of distinguishing, will become apparent. But if the viewer is considered to be nothing but a sighted mind,⁴ as though set in a point, not carrying with it any magnitudes either in reality or in the imagination, and considering in things only what may be 15R grasped by the intellect, like numbers, proportions, and angles, then no distinction will present itself. Therefore, these temples will be called similar, since they cannot be distinguished except by this co-observation, either among themselves or with a third thing, and not 20R at all when viewed individually and by themselves.

[6] This perspicuous and practical and general description of similarity will be beneficial to us for geometric demonstrations, as will soon become clear. We will say that two proposed figures are similar if nothing could be noted in the one, viewed individually, that could not 25R equally be apprehended in the other. And so it follows that the ratio or proportion of the constituents must be the same in both, otherwise a distinction would appear by themselves individually, i.e. even if no co-observation of both of them were arranged. But geometers, since they lacked a general notion of similarity, defined similar figures from equal corresponding angles, which is specific and does not reveal the nature of similarity itself in general. And so they had to take a circuitous route to demonstrate what is clear at the first glance from our notion. But let us come to examples. 30R

⁴Latin *mentem oculatam*, literally 'eye-ed mind'.

[7] Ostenditur in *Elementis*, Triangula similia seu aequiangula latera habere proportionalia, et vicissim; sed hoc multis ambagibus Euclides quinto demum libro conficit, cum primo statim ostendere potuisset *Elemento*, si nostram notionem
 5 fuisset secutus. Demonstrabimus primum, *triangula aequiangula esse similia*.

[7] It is shown in the *Elements* that similar, or equiangular, triangles have proportional sides, and vice versa, but Euclid finally completes this in the fifth book by many detours, when he could have shown it immediately in the first *Element*, if he had followed our notion. We
 5R will show first that *equiangular triangles are similar*.

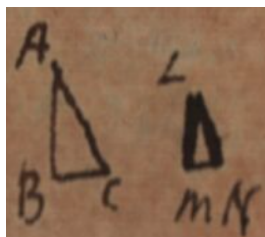


Figure 1

[8] Esto triangulum ABC , et aliud rursus LMN , sintque anguli A, B, C ipsis L, M, N respective aequales; dico triangula esse similia. Utor autem hoc *Axiomate novo*: *Quae ex determinantibus* (seu datis sufficientibus) *discerni non possunt, ea omnino discerni non posse*, cum ex determinantibus caetera omnia oriantur. Jam data basi BC , datisque angulis B , et C (adeoque et angulo A) datum est triangulum ABC ; itemque data basi MN , datisque angulis M, N , (adeoque et angulo
 15 L) datum est triangulum LMN . Sed ex his datis sufficientibus singulatim discerni triangula non possunt. Nam in uno quoque data sunt basis, et duo ad basin anguli; jam basis angulis conferri nequit; nihil aliud ergo superest quod in triangulo alterutro ex determinantibus sigillatim spectato
 20 examinari possit, quam ratio anguli cujusque dati ad rectum vel duos rectos, id est anguli ipsius magnitudo. Quae ipsa cum utrobique eadem reperiantur, necesse est triangula sigillatim discerni non posse, adeoque similia esse. Nam ut hoc in Scholii modum addam, etsi magnitudine triangula dis-

[8] Let there be a triangle ABC and again another LMN , and let the angles A, B, C be equal to L, M, N respectively; I claim that the triangles are similar. Moreover, I will use this new axiom: *whatever cannot be distinguished through the determiners* (i.e. sufficient givens) *cannot be distinguished at all*, since everything else arises from the determiners. Now, given base BC and given angles B and C (and so also angle A), the triangle ABC is given; likewise, given base MN , and given angles M, N (and so also angle L), the triangle LMN is given. But the triangles cannot be distinguished through these sufficient
 10R givens individually. For in each one, both the base and the two angles to the base are given. Now the base cannot be compared with the angles, therefore nothing else remains from the determiners which could be evaluated in either triangle viewed individually, other than the ratio of each given angle to a right angle or to two right angles, that is, the magnitude of the angle itself. Since these things are found to be the same, it is necessary that the triangles cannot be distinguished individually, and so they are similar. I might add as a
 15R sort of scholium that even if the triangles could be distinguished in
 20R

cerni possint, tamen magnitudo nisi per coobservationem vel triangulorum amborum simul, vel utriusque cum aliqua mensura, agnosci non potest, sed ita jam non tantum spectarentur singulatim, quod postulatur.

5 [9] Vicissim manifestum est, *Triangula similia etiam aequian-*
gula esse, alioqui si esset angulus aliquis ut A , in triangulo
 ABC , cui nullus reperiretur aequalis angulus in triangulo
 LMN , utique daretur angulus in ABC , habens rationem ad
10 quam non habet ullus in LMN , quod sufficit ad triangulum
 ABC , a triangulo LMN singulatim distinguendum. Constat
etiam *Triangula similia habere latera proportionalia*. Nam
si dentur duo aliqua latera, velut AB , BC , habentia rationem
inter se, quam nulla trianguli LMN latera inter se habeant,
15 jam poterit alterum triangulum ab altero singulatim discerni.
Denique *si latera proportionalia sint, triangula similia erunt*.
Quoniam enim datis lateribus data sunt triangula, sufficit
(per axioma nostrum) ex laterum ratione discrimen haberi
non posse, ut ex nullo in Triangulis his singulatim spectatis
20 alio haberi posse judicemus. Ex his vero etiam patet, *Tri-*
angula aequiangula habere latera proportionalia, et vicissim.

Eodem modo primo statim Mentis obtutu ex nostra simili-
tudinis notione directe ostenditur, *circulos esse ut quadrata*
diametrorum, quod Euclides demum decimo libro ostendit,
25 et quidem per inscripta et circumscripta, rem reducendo ad
absurdum, cum tamen nullis ambagibus esset opus.

[10] Diametro AB descriptus sit circulus, eique circumscrip-
tum diametri quadratum CD : Eodemque modo diametro LM
descriptus sit circulus eique circumscriptum diametri quadra-
30 tum NO . Determinatio utrobique est similis, circulus circulo,
quadratum quadrato, et accommodatio quadrati ad circulum,

magnitude, nonetheless the magnitude cannot be recognized except through co-observation either of both triangles simultaneously, or of each one with some measure, but then they would not be viewed just individually as was required.

5R [9] Conversely, it is clear that *similar triangles are also equiangular*;
otherwise, if there were some angle, say A , in triangle ABC for which
no equal angle could be found in triangle LMN , then there would
certainly be an angle in ABC having a ratio to two right angles (or to
the sum of all angles of the triangle) which none has in LMN , and this
10R suffices to distinguish triangle ABC from triangle LMN individually.
It is also established that *similar triangles have proportional sides*. For
if some two sides were given, such as AB and BC , having a ratio to each
other that no sides of triangle LMN have to each other, then the one
triangle could be distinguished from the other individually. Finally, *if*
15R *the sides are proportional, the triangles will be similar*. Indeed, since
triangles are given when the sides are given, it suffices (by our axiom)
that a distinction could not be obtained from the ratio of the sides
in order for us to judge that it could not be obtained from anything
else in these triangles viewed individually. But from these things it is
20R also clear that *equiangular triangles have proportional sides, and vice*
versa.

In the same way, right at first mental glance, one shows directly from
our notion of similarity that *circles are as the squares of the diameters*,
something Euclid shows finally in book ten and indeed by means of
25R inscribed and circumscribed [figures], reducing the matter to absurdity,
when actually no detours were necessary.

[10] Let a circle be described with diameter AB , and let the square CD ⁵
of the diameter be circumscribed on it; and in the same way let a circle
be described with diameter LM , and let the square NO of the diameter
30R be circumscribed on it. The determination of both is similar, the circle
to the circle, the square to the square, and the fitting of the square

⁵Meaning: the square whose side is the diameter CD .

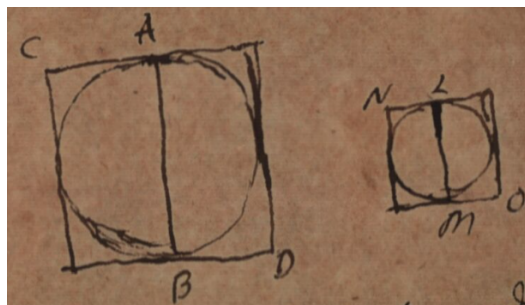


Figure 2

itaque (per Axioma supradictum) figurae $ABCD$, et $LMNO$ sunt similes. Ergo (per definitionem similitudinis) erit circulus AB ad quadratum CD , ut circulus LM ad quadratum NO , ergo etiam circulus AB ad circulum LM , est ut quadratum CD ad quadratum NO . Quod affirmabatur. Pari ratione *sphaerae* ostendentur esse *ut cubi diametrorum*. Et in universum in similibus, lineae, superficies, solida, homologa erunt respective ut longitudines, quadrata, cubi laterum homologorum. Quod hactenus generaliter assumptum magis quam demonstratum est.

to the circle, and so (by the aforementioned axiom) the figures $ABCD$ and $LMNO$ are similar. Therefore (by the definition of similarity) the circle AB is to the square CD as the circle LM is to the square NO ; therefore also the circle AB is to the circle LM as the square CD is to the square NO , as we claimed. By comparable reasoning, *spheres* will be shown to be *as the cubes of the diameters*. And in similars in general, homologous curves, surfaces, and solids will be as the lengths, squares, and cubes of homologous sides, respectively. Something which has until now generally been assumed rather than demonstrated.

[11] Porro haec consideratio, quae tantam praebebat facilitatem demonstrandi veritates alia ratione difficulter demonstrandas; etiam novum calculi genus nobis aperuit, a calculo Algebraico toto coelo diversum, notisque pariter, et usu notarum operationibusve novum. Itaque Analysin situs appellare placet, quod ea situm recta et immediate explicat; ita ut figurae etiam non delineatae per notas in animo depingantur, et quicquid ex figuris imaginatio intelligit empirica, id ex notis calculus certa demonstratione derivet, caeteraque etiam omnia consequatur, ad quae imaginandi vis pertingere non potest. Imaginationis ergo supplementum, et ut ita dicam perfectio in hoc, quem proposui, calculo situs continetur, neque

[11] This consideration, furthermore, which affords such ease in demonstrating truths that would be difficult to demonstrate by another method, has also opened to us a new kind of calculus, completely different from the algebraic calculus, equally new in its signs and in its use of signs, or its operations. And so I like to call it the analysis of situation, because it unfolds [*explicat*] situation directly and immediately, so that figures might be depicted in the mind through the signs even without being drawn, and whatever the imagination understands empirically from figures, the calculus will derive from the signs by sure demonstration, and even obtain everything else which the power of imagining cannot reach. Therefore a supplement to, and if I may say so the perfection of, the imagination is contained in this calculus

tantum ad Geometriam, sed etiam ad Machinarum inventiones, ipsasque machinarum naturae descriptiones usum hactenus incognitum habebit.

of situation I have proposed, and it will have uses never known until now, not only in geometry, but also for the invention of machines and the descriptions of nature's own machines.⁶

⁶The Latin *ipsasque machinarum naturae descriptiones* can be parsed either as 'and the descriptions themselves of the nature of machines', or as we have done. Note that a prior, cancelled draft had 'explicationesque', apparently instead of this phrase.